

DHAMMIKO BODHI AVATARA

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PROFILE

Computer Science graduate from University of Tanjungpura (GPA: 3.74/4.00) with 2+ years of specialized experience in Python Backend, Computer Vision (OpenCV), and IoT Embedded Systems. Proven track record in architecting systems that seamlessly integrate hardware data (ESP32/Raspberry Pi) into web backends. Experienced in leading robotic teams, competing in prestigious national competitions (Gemastik IoT), and securing national research grants (PKM-RE).

EDUCATION

Universitas Tanjungpura – S1 Informatika (GPA: 3.74) | Pontianak, Kalimantan Barat (2021 - 2025)

PROJECTS & ORGANIZATIONAL EXPERIENCE

Undergraduate Thesis Project | Backend & Computer Vision Engineer (Dec 2024 - Oct 2025)

- Engineered a RESTful API backend service using Python (FastAPI) and Docker under an Agile SDLC framework to automate official passport photo format verification.
- Developed a core computer vision pipeline utilizing OpenCV for face detection (Haar Cascade), background extraction (HSV color space), head-pose evaluation (centroid analysis), and formal attire verification (cosine similarity).
- Achieved a system reliability rate exceeding 90% under standard testing scenarios, with high-performance processing times optimized between 2–5 seconds per image.

Program Kreativitas Mahasiswa Riset Eksakta (PKM-RE) | *Embedded Systems Engineer* (2024)

- Secured national research grant funding from Belmawa DIKTI for the project "Pemodelan Sistem dalam Analisis Rasa Buah Jeruk Pontianak Berbantuan Raspberry Pi Camera."
- Spearheaded hardware design and electrical architecture, including wiring installation and sensor calibration on Raspberry Pi to ensure device operational stability.
- Developed embedded software integrations to seamlessly stream data inputs from the Raspberry Pi Camera into the computational analysis model.

Gemastik XVII | *Embedded Systems Engineer – Envirowatchers Team* (2024)

- Competed in the Intelligent Device division by designing "Envirowatchers," an IoT-based smart embedded system for real-time air quality monitoring.
- Owned the hardware configuration and sensor calibration processes using ESP32 microcontrollers to guarantee accurate environmental data acquisition.
- Formulated a wireless data transmission pipeline using Wi-Fi communication protocols to stream continuous, stable sensor telemetry to a web backend.

Tim Robot FT UNTAN – Team Lead & Core Programmer KRSBI-B (2022 - 2024)

- Led cross-functional team coordination and integrated software-hardware modules to optimize wheeled robot performance for the National Wheeled Soccer Robot Competition (KRSBI-B).
- Built a desktop base station application to coordinate and transmit real-time control instructions to multiple wheeled robots over a local Wi-Fi network.
- Implemented real-time computer vision (OpenCV) systems on wheeled robots for object detection and tracking, while enforcing code management standards via GitHub.

SKILLS

- Programming Languages** : Python, JavaScript, C++.
- Framework & Libraries**: OpenCV, FastAPI, ROS2, React, Tailwindcss.
- Tools & DevOps**: Visual Studio Code, Git, Github, Docker, MySQL.
- Languages**: Indonesian (Native), English (Professional Working Proficiency), Japanese (Elementary).

CERTIFICATIONS

- Programmer* – Badan Nasional Sertifikasi Profesi (BNSP), 2025.